Lesson 7 — Tables

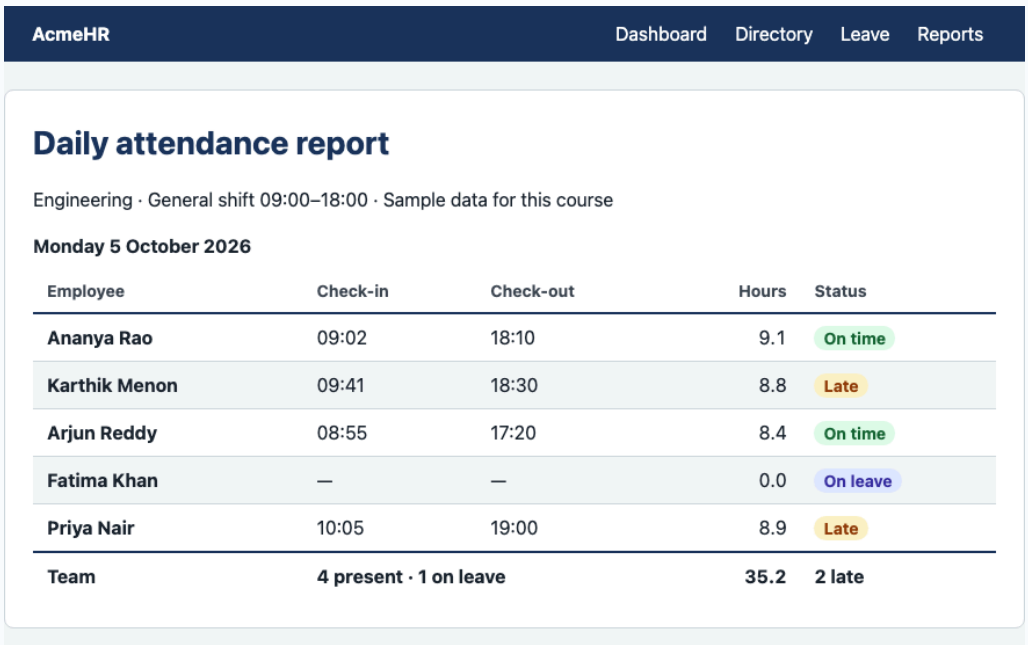
Course: HTML & CSS for Full Stack · **Session 7 of 9** · Learning path · **Previous:** Lesson 6 · **Next:** Lesson 8 — Responsive Design

> **Module Outcome:** You can build a data table that screen readers understand, style it so numbers are easy to compare, and keep it usable on a phone.

You will build	The AcmeHR **Daily attendance report**
New HTML	`table`, `caption`, `thead`, `tbody`, `tfoot`, `tr`, `th` (`scope`), `td`, `colspan`, `rowspan`
New CSS	`border-collapse`, `font-variant-numeric`, `white-space`, `overflow-x`, the `:nth-child()` pseudo-class
You need	Lessons 1–6

From the requirements. AcmeHR records each employee's check-in and check-out. Attendance reports are daily, monthly and exception-based, and late arrivals are reported.

Where we are going:



All example files: `examples/lesson07/`

> **Remember:** Tables are for **data** — rows and columns that belong together, like a spreadsheet. Never use a table to lay out a page; that is what Grid and Flexbox are for.

Part A — Table HTML

1. Rows and cells

```
<table>
  <tr>
    <th>Employee</th>
    <th>Check-in</th>
    <th>Check-out</th>
  </tr>
  <tr>
    <td>Ananya Rao</td>
    <td>09:02</td>
    <td>18:10</td>
  </tr>
  <tr>
    <td>Karthik Menon</td>
    <td>09:41</td>
    <td>18:30</td>
  </tr>
</table>
```

Output:

Employee	Check-in	Check-out
Ananya Rao	09:02	18:10
Karthik Menon	09:41	18:30

Element	Means
`<table>`	The table
`<tr>`	**T**able **r**ow
`<th>`	**T**able **h**eader cell — a label (bold and centred by default)
`<td>`	**T**able **d**ata cell — a value

2. Caption, sections, and `scope`

```
<table>
  <caption>Attendance, Monday 5 October 2026</caption>
  <thead>
    <tr>
      <th scope="col">Employee</th>
      <th scope="col">Check-in</th>
      <th scope="col">Check-out</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">Ananya Rao</th>
      <td>09:02</td>
```

```
      <td>18:10</td>
    </tr>
    <tr>
      <th scope="row">Karthik Menon</th>
      <td>09:41</td>
      <td>18:30</td>
    </tr>
  </tbody>
</table>
```

Output:

Attendance, Monday 5 October 2026

Employee	Check-in	Check-out
Ananya Rao	09:02	18:10
Karthik Menon	09:41	18:30

- `<caption>` is the table's title. Screen readers announce it first.
- `<thead>` holds the header rows; `<tbody>` holds the data rows.
- `scope="col"` — this header labels the **column** below it. `scope="row"` — this header labels the **row** to its right.

Why `scope` matters: A screen reader in the cell "09:41" can say "Karthik Menon, Check-in, 09:41". Without header cells and `scope`, it only says "09:41".

3. A total row — `tfoot`

```
<table>
  <caption>Leave taken in September 2026</caption>
  <thead>
    <tr>
      <th scope="col">Employee</th>
      <th scope="col">Casual</th>
      <th scope="col">Sick</th>
      <th scope="col">Total</th>
    </tr>
  </thead>
  <tbody>
    <tr><th scope="row">Ananya Rao</th><td>1</td><td>0</td><td>1</td></tr>
    <tr><th scope="row">Karthik Menon</th><td>2</td><td>1</td><td>3</td></tr>
    <tr><th scope="row">Meera Iyer</th><td>0</td><td>2</td><td>2</td></tr>
  </tbody>
  <tfoot>
    <tr><th scope="row">Team total</th><td>3</td><td>3</td><td>6</td></tr>
  </tfoot>
</table>
```

Output:

Leave taken in September 2026

Employee	Casual	Sick	Total
Ananya Rao	1	0	1
Karthik Menon	2	1	3
Meera Iyer	0	2	2
Team total	3	3	6

`<tfoot>` holds summary rows: totals, averages, counts.

4. Cells that span — `colspan` and `rowspan`

```
<table>
  <caption>Shift and actual times</caption>
  <thead>
    <tr>
      <th scope="col" rowspan="2">Employee</th>
      <th scope="col" colspan="2">Shift</th>
      <th scope="col" colspan="2">Actual</th>
    </tr>
    <tr>
      <th scope="col">Start</th>
      <th scope="col">End</th>
      <th scope="col">In</th>
      <th scope="col">Out</th>
    </tr>
  </thead>
  <tbody>
    <tr><th scope="row">Ananya Rao</th><td>09:00</td><td>18:00</td><td>09:02</td><td>18:10</td></tr>
    <tr><th scope="row">Karthik Menon</th><td>09:30</td><td>18:30</td><td>09:41</td><td>18:30</td></tr>
  </tbody>
</table>
```

```
th, td { border: 1px solid #6b7280; padding: 4px 8px; }
```

Output:

Shift and actual times

Employee	Shift		Actual	
	Start	End	In	Out
Ananya Rao	09:00	18:00	09:02	18:10
Karthik Menon	09:30	18:30	09:41	18:30

- `colspan="2"` — the cell is two columns wide.
- `rowspan="2"` — the cell is two rows tall.

Common Error: A row with the wrong number of cells. Count them: with spans included, every row must cover the same number of columns, or the cells shift sideways.

Part B — Styling tables

5. `border-collapse`

```
td { border: 1px solid #1a365d; padding: 4px 8px; }
.separate { border-collapse: separate; }
.collapse { border-collapse: collapse; }
```

Output:



By default every cell has its own border with a gap between them. `border-collapse: collapse` merges them into single lines. Almost every styled table uses it.

6. A readable data table

```
table {
  border-collapse: collapse;   /* one line between cells, not two */
  width: 100%;
}
caption {
  text-align: left;
  font-weight: bold;
  padding-bottom: 8px;
}
th, td {
  padding: 8px 12px;
  border-bottom: 1px solid #d0d7de;
  text-align: left;
}
thead th {
  border-bottom: 2px solid #1a365d;
}
.num {
  text-align: right;
  font-variant-numeric: tabular-nums;   /* every digit the same width */
}
tbody tr:nth-child(even) {
  background-color: #f4f6f8;            /* zebra stripes */
}
```

Output:

Attendance, Monday 5 October 2026

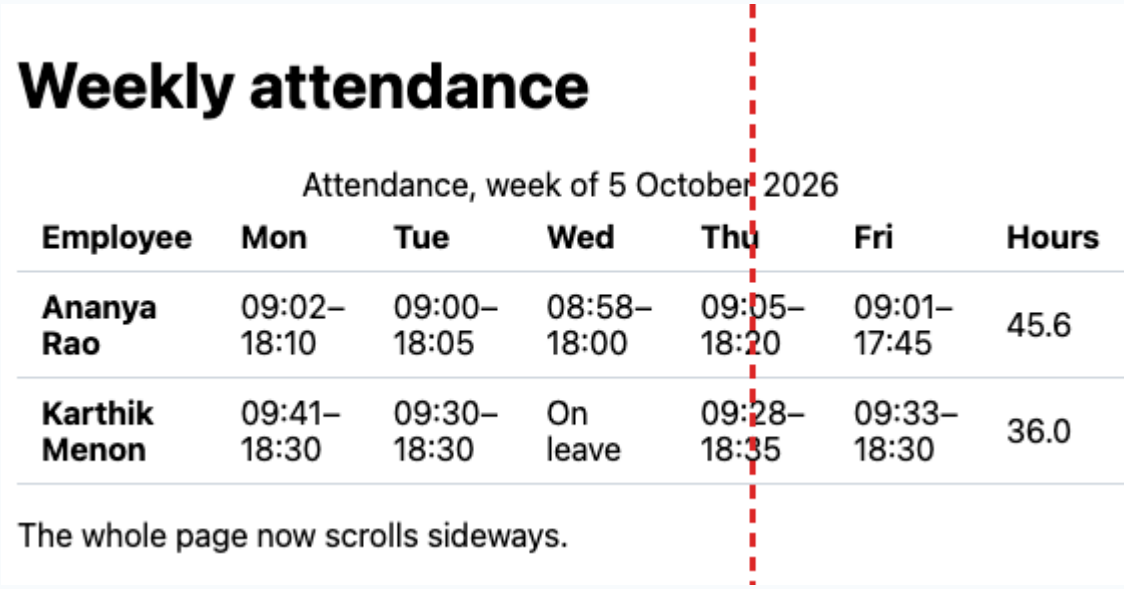
Employee	Check-in	Check-out	Hours
Ananya Rao	09:02	18:10	9.1
Karthik Menon	09:41	18:30	8.8
Meera Iyer	08:55	17:20	8.4
Rahul Sharma	10:05	19:00	8.9

Rule	Why
Text left, **numbers right**	Units, tens and decimal points line up, so values are easy to compare. Right-align the number's header too.
`font-variant-numeric: tabular-nums`	In many fonts "1" is narrower than "8". This makes every digit the same width, so columns of numbers line up exactly.
Lines between rows, not around every cell	Less visual noise
`tr:nth-child(even)`	A **pseudo-class** that picks every 2nd row (2, 4, 6…). `odd` picks 1, 3, 5. Stripes help the eye follow long rows.

7. Wide tables on a phone

A weekly table is wider than a phone screen. Without help, the **whole page** scrolls sideways:

Output at 375px — the red dashed line is the edge of the phone screen:



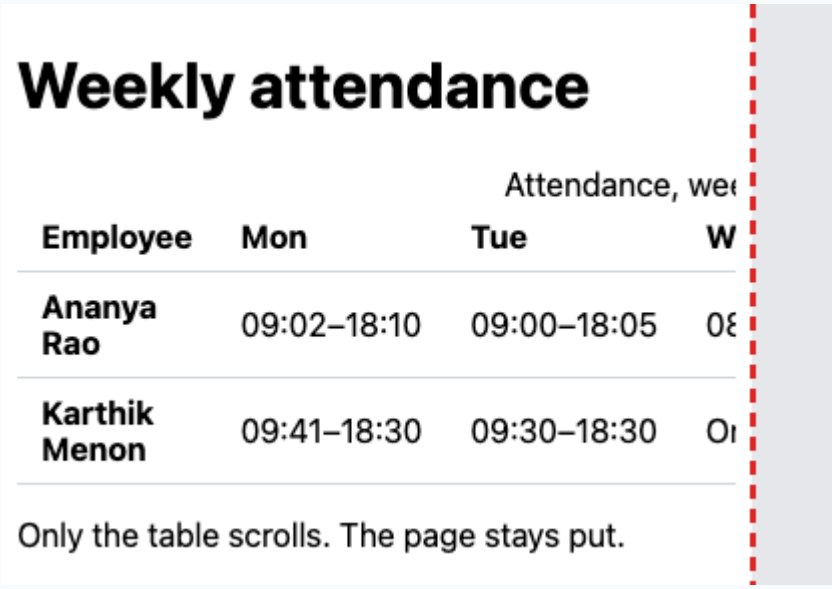
Wrap the table in a `div` that scrolls by itself:

```
<div class="table-scroll">
  <table>
```

```
    ...
  </table>
</div>
```

```
.table-scroll {
  overflow-x: auto;      /* only this box scrolls sideways */
}
td {
  white-space: nowrap;   /* keep each time range on one line */
}
```

Output at 375px:



- `overflow-x: auto` — if the content is wider than the box, the box gets its own sideways scroll.
- `white-space: nowrap` — text in the cell never breaks onto a second line.

The page stays put; only the table scrolls. Every reporting screen needs this.

8. Putting it together — the attendance report

The HTML:

```
<main>
  <h1>Daily attendance report</h1>
  <p>Engineering · General shift 09:00–18:00 · Sample data for this course</p>

  <div class="table-scroll">
    <table>
      <caption>Monday 5 October 2026</caption>
      <thead>
        <tr>
          <th scope="col">Employee</th>
          <th scope="col">Check-in</th>
          <th scope="col">Check-out</th>
          <th scope="col" class="num">Hours</th>
          <th scope="col">Status</th>
        </tr>
      </thead>
      <tbody>
        <tr><th scope="row">Ananya Rao</th><td>09:02</td><td>18:10</td><td class="num">9.1</td><td><spa
```

```
        <tr><th scope="row">Karthik Menon</th><td>09:41</td><td>18:30</td><td class="num">8.8</td><td><
        <tr><th scope="row">Arjun Reddy</th><td>08:55</td><td>17:20</td><td class="num">8.4</td><td><spa
        <tr><th scope="row">Fatima Khan</th><td>—</td><td>—</td><td class="num">0.0</td><td><span class=
        <tr><th scope="row">Priya Nair</th><td>10:05</td><td>19:00</td><td class="num">8.9</td><td><spa
      </tbody>
      <tfoot>
        <tr><th scope="row">Team</th><td colspan="2">4 present · 1 on leave</td><td class="num">35.2</t
      </tfoot>
    </table>
  </div>
</main>
```

(The page `header` and `nav` are the same as in Lessons 5 and 6.)

Output, HTML only:

AcmeHR

- Dashboard
- Directory
- Leave
- Reports

Daily attendance report

Engineering · General shift 09:00–18:00 · Sample data for this course

Monday 5 October 2026

Employee	Check-in	Check-out	Hours	Status
Ananya Rao	09:02	18:10	9.1	On time
Karthik Menon	09:41	18:30	8.8	Late
Arjun Reddy	08:55	17:20	8.4	On time
Fatima Khan	—	—	0.0	On leave
Priya Nair	10:05	19:00	8.9	Late
Team	4 present · 1 on leave		35.2	2 late

The table part of `report.css`:

```
/* Only the table scrolls on a narrow screen, not the page */
.table-scroll {
  overflow-x: auto;
}

table {
  width: 100%;
  border-collapse: collapse;
}

caption {
  padding-bottom: 8px;
  text-align: left;
  font-weight: bold;
}

th,
td {
  padding: 8px 12px;
  border-bottom: 1px solid #d0d7de;
  text-align: left;
  white-space: nowrap;
}
```

```
thead th {
  border-bottom: 2px solid #1a365d;
  font-size: 0.875rem;
  color: #3d4654;
}

tbody tr:nth-child(even) {
  background-color: #f4f6f8;
}

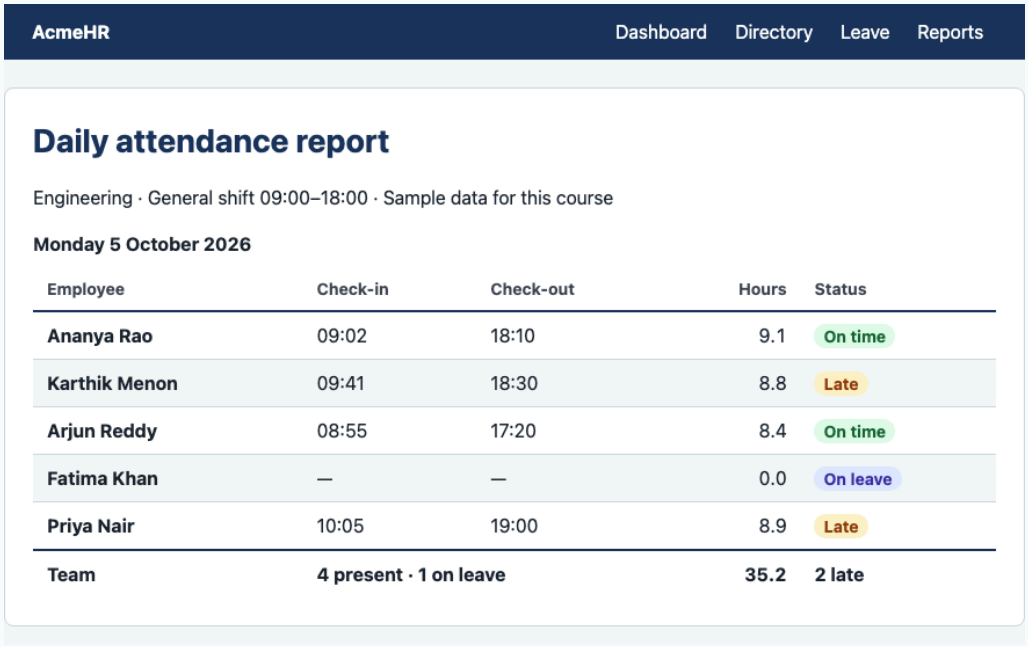
tfoot th,
tfoot td {
  border-top: 2px solid #1a365d;
  border-bottom: 0;
  font-weight: bold;
}

/* Numbers: right-aligned, every digit the same width */
.num {
  text-align: right;
  font-variant-numeric: tabular-nums;
}

/* Status: text + colour, never colour alone */
.status {
  padding: 2px 8px;
  border-radius: 999px;
  font-size: 0.875rem;
  font-weight: bold;
}

.on-time { background-color: #dcfce7; color: #166534; }
.late    { background-color: #fef3c7; color: #92400e; }
.leave   { background-color: #e0e7ff; color: #3730a3; }
```

Output, styled:



Files: `10_report_styled.html` · `report.css`

- The **status** is written as a word ("Late") *and* coloured. A colour-blind manager can still read it.

- `.num` is on the header cell too, so "Hours" sits right above its numbers.
- The `tfoot` row uses `colspan="2"` for a sentence that covers two columns.

> **Where used?** Report rows come from the database: the template loops over the records and writes one `tr` per employee (`{% for row in rows %}` in Django, `th:each` in Thymeleaf). Reports often also offer a "Download CSV" link that returns the same data as a file.

Try these

1. **(Easy)** Add a sixth employee, Rahul Sharma: 09:00–18:00, 9.0 hours, On time. Update the footer.
2. **(Easy)** Change the zebra stripes to the **odd** rows. Which looks better with the header line?
3. **(Medium)** Add a "Shift" column between Employee and Check-in. Update every row, including `tfoot`.
4. **(Medium)** Remove `white-space: nowrap` and narrow your window. What happens to the times and the status pills?
5. **(Stretch)** Build a *monthly* summary table: one row per employee, columns Present, Late, On leave, Absent (all right-aligned numbers), and a `tfoot` with team totals.

> **Interview:** "*When should you use a table?*" — Only for tabular data — values with a row and a column meaning. Never for page layout.

> **Interview:** "*How do you make a table accessible?*" — A `caption`, header cells with `th`, and `scope="col"` or `scope="row"` so each data cell is linked to its headers. Do not rely on colour alone for status.

> **Interview:** "*How do you handle a wide table on mobile?*" — Wrap it in a container with `overflow-x: auto`, so only the table scrolls horizontally, not the whole page.

> **Key takeaways**
> - `table` → `caption`, `thead`, `tbody`, `tfoot` → `tr` → `th` / `td`.
> - Header cells use `th` with `scope="col"` or `scope="row"`.
> - `colspan` / `rowspan` merge cells — every row must still cover the same columns.
> - `border-collapse: collapse`, lines between rows, numbers right-aligned with `tabular-nums`.
> - Wrap wide tables in `overflow-x: auto`.
> - Never show status with colour alone.

Lab

👉 Lab 07 — AcmeHR attendance report

Next: Lesson 8 — Responsive Design